The alternative hypothesis in (i) is known as two-sided alternative and the alternative hypothesis in (ii) is known as one-sided (right) alternative and (iii) is known as one-sided (left) alternative.

1.5 ERRORS IN STATISTICAL HYPOTHESES TESTING

A **statistical decision** in a hypotheses testing problem is either of rejecting or not rejecting H_0 based on a given random sample. Statistical decisions are governed by certain rules, developed by applying a statistical theory, which are known as **decision rules**. The decision rule leading to rejection of H_0 is called as **rejection rule**.

The null hypothesis may be either true or false, in reality. Under this circumstance, there will arise four possible situations in each hypotheses testing or decision making problem as displayed in Table 1.5.

Table 1.5 Decision Table

	H_0 is true	H_0 is false
Reject H_0	Type I error	Correct decision
Do not Reject H_0	Correct decision	Type II error

It must be recognized that the final decision of rejecting H_0 or not rejecting H_0 may be incorrect. The error committed by rejecting H_0, when H_0 is really true, is called **type I error**. The error committed by not rejecting H_0, when H_0 is false, is called **type II error**.

Example 1.3

A soft drink manufacturing company makes a new kind of soft drink. Daily sales of the new soft drink, in a city, is assumed to be distributed with mean sales of ₹40,000 and standard deviation of ₹2,500 per day. The Advertising Manager of the company considers placing advertisements in local TV Channels. He does this on 10 random days and tests to see whether or not sales has increased. Formulate suitable null and alternative hypotheses. What would be type I and type II errors?

Solution:

The Advertising Manager is testing whether or not sales increased more than ₹40,000. Let μ be the average amount of sales, if the advertisement does appear.

The null and alternative hypotheses can be framed based on the given information as follows:

Null hypothesis: H_o: $\mu = 40000$

i.e., The mean sales due to the advertisement is not significantly different from ₹40,000.

Alternative hypothesis: H_1: $\mu > 40000$

i.e., Increase in the mean sales due to the advertisement is significant.

(i) If type I error occurs, then it will be concluded as the advertisement has improved sales. But, really it is not.

(ii) If type II error occurs, then it will be concluded that the advertisement has not improved the sales. But, really, the advertisement has improved the sales.

12th_Statistics_EM_Unit_1.indd 8 1/28/2020 4:47:29 PM